

Requests

TTFT / TPOT SLO (ms)



Chatbot

100 / 150



Summary

800 / 100



Translation

3000 / 30

Deadline-Aware Request Dispatcher

Prefill Request

Arrival + TTFT SLO

Decode Request

Last token + TPOT SLO

Waiting

R₄

R₅

R₃

...

*Phase-Aware
Request Ordering*

*Deadline-Aware
Preemption*

Running

R₄

R₁

R₂

R₃

...

Step-Time-Model-Guided Token Allocator

Prefill

Decode

KV context

Attention

Decode deadlines

D₁

D₂

D₃

...

Step-time model ⌚

Predicted time

Time budget

Token Allocation

Token cap

Prefill

Decode

Blank

TPOT budget



vLLM Execution

Continuous batching



Better Multi-SLO Goodput